The students are very enthusiastic about using computers and often try to monopolise the machine! They are taught the aids and appliances necessary to operate the computer—starting with the typewriter to learn touch-typing, followed by the Perkins Brailler (a special Braille typewriter) and finally moving on to the computer keyboard.

The deaf-blind and the blind cannot see the monitor and the mouse serves no purpose either. This is where the electronic Braille display board comes in. This display board indicates each and every change that takes place on the monitor.

The deaf-blind and the blind-can understand these changes by moving their fingertips across the cells on the display board, which are in Braille.

The Braille cells move as data is entered through the keyboard enabling the deaf-blind to read the text on the screen, manoeuvre accordingly, make changes and even e-mail, surf and download content from the Internet.

The Computerised Mini Braille Press

An integral part of HKIDB is the pioneering Computerised Mini Braille Press project, set up in January 2002. Here, the deaf and deaf-blind are trained to use computers and undertake computer-related programming and designing. This computer-training unit-cum-mini braille press produces a variety of materials to suit the needs of the deaf-blind/blind/low vision/sighted and hearing individuals.

The deaf students are trained in graphic design and produce tactile graphic educational material. While, the blind, those who are proficient in Braille, help in proofreading. The newsletter *Deafblindness in Asia: A Communication Link* is composed and published by the Braille Press and is circulated among all centres for the deaf-blind in Asia, Europe and other parts of the world.

Vacha says, "The success of our Computerised Mini Braille Press is thanks to the contribution of the Rangoonwala Foundation, UK. We also owe a debt of gratitude to the British High Commission, Christoffel Blindenmission and other donors. We are equally grateful to our trainers for their efforts to teach the students."

The Process

A software called JAWS (Jobs Accessing With Speech) enables the blind (who have normal hearing) to use computers by listening to the audio instructions.

JAWS is also essential for synthesising text or commands on screen into Braille, which then appear on the electronic Braille display board. The deaf who have normal vision work on graphics using CorelDraw, while the printing and the packaging jobs are looked after by the deaf-blind.

Computer trainer Devyani P states, "We had nobody to emulate and had to start from scratch. We researched a lot to produce material in Braille and normal print. We started with greeting cards."

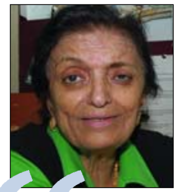
Tools And Techniques

The Duxbury Braille Translator (DBT) is a software that translates text, say, from MS Word, into Braille. Printouts in Braille can then be taken through an embosser. Even those who do not know Braille, can type in regular English and print in Braille, which otherwise would be time-consuming and tedious even for those who can write in Braille.

Picture In A Flash (PIAF) is a piece of hardware that produces excellent tactile graphics. However, the A3 paper required for PIAF is costlier as compared to the one used on the Tiger Advantage machine, another output device that prints tactile graphics. But the print-outs from the Tiger Advantage machine are no match for the PIAF ones that are more distinguishable and attractive.

Picture In A Flash or PIAF is used to print tactile graphics

The equipment used at the Braille press is expensive—a single electronic Braille display board costs an astronomical Rs 2.5 lakh! Prices of other equipment range from Rs 50,000 to Rs six lakh.



"Don't the deafblind have an equal right to be computer-savvy? Computer literacy enables them to lead normal lives

Beroz N Vacha
Director and Consultant, HKIDB, Mumbai

At The Workshop

Assistant computer trainer Pradip Sinha (25) is a father of five. His family is in Kolkata, while he stays in Mumbai, as the facilities provided by HKIDB are not available elsewhere.

A Devkumar (25) who is deaf, is the DTP operator. The hearing-impaired but extraordinarily talented Mahesh Joshi (31) and Dhirendra Dubey (27) are graphics designers. A first-year B.Com student, Aarti Shetty (20) is hearing-impaired and works as the data-entry operator. Naina (30) was diagnosed with juvenile diabetes and her vision and hearing is on the decline. She says, "In seven months I have learnt Braille, computers, candle and perfume making and other crafts. I am glad to be independent."

Incidentally, Dhale (27) is one of the people responsible for teaching Amitabh Bachchan manual communication for the deaf-blind for the film *Black*, where the Big B plays the teacher of a deaf-blind girl. Sanjay Leela Bhansali and his team visited HKIDB to study the characteristics of the deaf-blind.

Income Generation

Balaji J, administrator at the Braille Press, says, "Our assignments range from printing stationery, brochures to educational material. These assignments help us disburse stipends and salaries to the deaf and deaf-blind who work in the Braille press."

Their clientele reads Sunchem Corp, Sauradip Chemical Industries, Fine Stitches, Tankaria Exim, Shreepaul and Company and various other firms. They recently completed an assignment for the Mumbai District AIDS Control Society—a brochure in Hindi and English with 5,000 copies in normal print and 10,000 in Braille. They received a work order for 2,500 copies from Dignity Foundation. At present, they are working on a safety introduction manual in Braille for Jet Airways.

At exhibitions and displays of the Braille Press products, visitors are stunned to see the deaf and deaf-blind expertly handle computers. Through learning and using computers, the physically handicapped can become contributing members of the society and lead successful lives.

For more information, visit **www.helenkelleridb-mumbai.org**. ⊠

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